Metropolis-Hastings algorithm.

A distn π (target)

Want to generate r.v's from π.

$X \sim \pi$ $\quad \mu = EX$.

$X_1 \dots X_n \overset{iid}{\sim} \pi \qquad \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X} \xrightarrow[SLLN]{a.s.} \mu$

Cannot generate from π either directly or indirectly

- Find a Markov chain with transition matrix P which satisfies $\pi P = \pi$
- Generate a Markov chain $Y_1, Y_2, \dots Y_n$ from P
- Under certain conditions $\frac{1}{n}\sum_{i=1}^{n} Y_i \to \mu$.

$x, y \to$ states of Markov chain

$P = ((p_{x,y})) \qquad p_{x,y} = \left(\frac{\omega(y)}{\omega(x)} \wedge 1\right)/d$.

Do not know π. But know it upto a const i.e.

$\pi(x) = c\,\omega(x)$.

① $d \geq$ max number of states that you can go to in 1 step from any x.

$d = m(m-1)k(k-1)$

② Restrict to $\{x : \pi(x) > 0\}$

③ What if the $\begin{bmatrix} \overset{+1}{(i,j)} & \overset{-1}{(i,j')} \\ \underset{-1}{(i',j)} & \underset{+1}{(i',j')} \end{bmatrix}$ (i,j')th element of x is zero?

Then stay at x.

Choose i, i', j, j' again from $m \times k$ table.

$x \to y$

Proposition: The transition probability matrix

$$p(x,y) = \begin{cases} \left(\frac{\omega(y)}{\omega(x)} \wedge 1\right)/d & y \neq x,\ y \in T \\ 1 - \sum_{y \neq x} p(x,y) & y = x \end{cases}$$

is aperiodic and irreducible on T (for the eg.) and its unique stationary distribution is π.

Pf: π is a stationary distn of P.

We have shown this in 2 steps

(i) P satisfies the detailed balance eqns

$\pi(x)p(x,y) = \pi(y)p(y,x) \quad \forall x, y$.

(ii) If P satisfies detailed balance eqn in π then it satisfies $\pi P = \pi$.

For this eg: Aperiodic. There is some $x \in T$ so that $p(x,x) > 0$

Pick x that has one element 0.

If (i,j') combination is that element then algorithm dictates ~~stay~~ stay at x. So $p(x,x) > 0$.

Irreducible: You can reach any state y from any other state x. (Tutorial)

General result: Aperiodic & irreducible $P \Rightarrow \pi$ is unique.

Thm: Let $\{Y_n\}$ be aperiodic, irreducible, positive recurrent Markov chain P on a finite or countable state space $\mathcal{X}$ with stationary distn π and let f be a real valued fn on $\mathcal{X}$ such that $\sum_{x \in \mathcal{X}} |f(x)|\pi(x) < \infty$, then for any initial state x, with P^x probability one,

$$\lim_{n\to\infty} \frac{1}{n}\sum_{j=1}^{n} f(Y_j) = \mu := \sum_{x \in \mathcal{X}} f(x)\pi(x)$$

$X_1 \dots X_n \sim \pi$

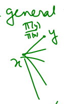
$\frac{1}{n}\sum_{i=1}^{n} f(X_i) \xrightarrow[SLLN]{} \mu$

where $Y_1, Y_2, \dots$ is a sequence from P.

Requires Ergodic Thm.

Modifications. d was crucial.

For a general state space $\mathcal{X}$, d may be infinite.



Instead of this $p(x,y) \propto \frac{\omega(y)}{\omega(x)}$ we want a diff p that leads to faster convergence.

$$p(x,y) = \begin{cases} \left(\frac{\pi_y}{\pi_x} \cdot \frac{q(y,x)}{q(x,y)} \wedge 1\right) q(x,y) & y \neq x \\ 1 - \sum_{y \neq x} p(x,y) & y = x. \end{cases}$$

where $Q = ((q_{xy}))$ "proposal" is some transition matrix.

① $((p))_{xy} = P$ is a transition matrix

② P satisfies detailed balance eqns.

③ $\pi P = \pi$.

Prev defn was special case of this with $q(x,y) = 1/d$

Continuous case: $\mathcal{X}$ is ~~count~~ uncountable and π is density on $\mathcal{X}$.

~~Let~~ Let Q be a transition fn s.t. for every $x \in \mathcal{X}$, Q has a density $q(x, \cdot)$, define transition density as

$$p(x,y) = \left(\frac{\pi(y)}{\pi(x)} \cdot \frac{q(y,x)}{q(x,y)} \wedge 1\right) q(x,y)$$

Choice of q. In Metropolis (1953), Q was chosen to be a symmetric matrix/density. Eg. for density Q could be normal

$q(x,y)$	$q(y,x)$
$Y \sim N(x,1)$	$X \sim N(y,1)$
$\frac{1}{\sqrt{2\pi}}\exp\{-\frac{1}{2}(y-x)^2\}$	$\frac{1}{\sqrt{2\pi}}\exp\{-\frac{1}{2}(x-y)^2\}$.

Some papers choose $q(x,y) = q(|x-y|)$

Normal also satisfies this.

Scale is important. Determines how quickly the chain moves / mixes / covers the whole space.

Very large scale may lead to convergence while a small scale will move in the same neighborhood.